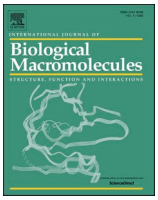
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A designed $ZrOCl_2$/ethylene glycol deep eutectic solvent for efficient lignocellulose valorization

Yunhua Bai, Xiong-Fei Zhang *, Mengjiao Yu, Jianfeng Yao *

Jiangsu Co-Innovation Center of Efficient Processing and Utilization of Forest Resources, College of Chemical Engineering, Nanjing Forestry University, Nanjing 210037, China

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ABSTRACT

Deep eutectic solvents (DESs) hold great potential in biorefining because they can efficiently deconstruct the recalcitrant structure of lignocellulose. In particular, inorganic salts with Lewis acids have been proven to be effective at cleaving lignin–carbohydrate complexes. Herein, a Zr-based DES system composed of metal chloride hydrate ($ZrOCl_2{\cdot}8H_2O$) and ethylene glycol (EG) was designed and used for poplar powder pretreatment. Zr^{4+}-based salts provide sufficient acidity for lignocellulose depolymerization. The acidity of the DES was analysed by the Kamlet-Taft solvatochromic parameter, and the results demonstrated that the acidity can be regulated by the DES composition. Under the optimum conditions ($ZrOCl_2{\cdot}8H_2O$:EG molar ratio of 1:2), the DES pretreatment removes nearly 100 % hemicellulose and 94.7 % lignin. The recovered lignin exhibited a low polydispersity of 1.7. The cellulose residues deliver an efficiency of 94.4 % upon enzymatic digestion. Moreover, the DES can be easily recovered with high yield and purity, and the recycled DES still maintains high delignification and enzymatic hydrolysis efficiencies. The proposed DES pretreatment technology is promising for biomass valorization.

1. Introduction

Owing to the scarcity of resources and the depletion of energy reserves, biomass energy has garnered widespread attention [1,2]. Notably, lignocellulose has emerged as a remarkable substitute for fossil fuels owing to its abundant sources, high yield, and inherent renewability [3–5]. For example, poplar sawdust is a processing waste from the wood manufacturing industry and a rich resource for biorefining. However, the complex structure of lignocellulose, known as biomass obstinacy, poses a significant challenge to realizing its full potential for high-value applications [6–8]. Efficient pretreatment technologies are urgently needed to achieve more efficient degradation and utilization of lignocellulose components [9,10]. Currently, common pretreatment methods primarily involve the use of acids, alkalis, organic solvents, and other chemical solvents [11,12].

Among these lignocellulosic pretreatment routes, deep eutectic solvents (DESs) have attracted considerable attention due to their low cost, environmental friendliness and biocompatibility with enzymes [13,14]. Typically, DES is regarded as a eutectic mixture of two or more compounds containing a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA) [10,15]. In general, the DESs used for biomass pretreatment are mainly based on quaternary ammonium salts as HBAs and acids/alcohols as HBD, where the acidity of DES is closely related to the fractionation effect of lignocellulose [16,17]. To modulate the acidity of DESs, inorganic salts are introduced into the DES system as the third component. This addition facilitates the disruption of lignin-carbohydrate complexes (LCCs). For example, Xia et al. introduced the metal salt aluminum chloride ($AlCl_3$) into the choline chloride (ChCl)/glycerol system. The inclusion of this inorganic salt elegantly modulated both the coordination states and the acidity strength of the DES [11]. Under the optimal pretreatment conditions, the lignin fractionation efficiency reached 95.5 %, with a high purity of 94.4 %. In another parallel effort, Wang et al. introduced ferric chloride ($FeCl_3$) into a ChCl/glycerol system, resulting in significant fractional removal of lignin (78.9 %). Moreover, the treated substrate exhibited a high retention of cellulose (95.2 %) and remarkable enzymatic hydrolysis efficiency (99.5 %) [18]. Very recently, DESs with metal salts as HBA have become an ideal solvent for lignocellulose degradation [19]. In our previous work, we designed a DES with a Lewis acid (consisting of $ZnCl_2$: lactic acid) and used it for lignin extraction from poplar sawdust. A DES with an optimized acidity resulted in 95.2 % lignin recovery and improved the enzymatic efficiency of cellulose [20]. The incorporation of metal salts

* Corresponding author.
E-mail addresses: zxf1990@njfu.edu.cn (X.-F. Zhang), jfyao@njfu.edu.cn (J. Yao).

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has enabled DESs to show great potential in extracting lignin from biomass. Additionally, polyols like ethylene glycol are used as HBD in DESs because of their multiple free hydroxyl groups. These groups occupy hydrogen bonding sites in lignocellulose, weakening interactions between lignocellulose components and interactions with the hydroxyl groups of lignin. This interaction facilitates efficient lignin separation and enhances cellulose accessibility to cellulases [21].

Based on the above studies, this work designed a Zr-based DES system composed of $ZrOCl_2{\cdot}8H_2O$ and ethylene glycol (EG). Zr^{4+}-based metal chloride provides sufficient Lewis acidity for lignocellulose processing [22], and the existence of EG is expected to tune the DES acidity and enhance the lignin isolation efficiency, thus facilitating the binding interaction between isolated cellulose and cellulase in subsequent enzymatic hydrolysis [23]. Poplar sawdust was selected as a substrate for pretreatment, and the pretreatment process is summarized in Scheme 1. The results showed that the $ZrOCl_2$-EG DES could efficiently separate hemicellulose and lignin while retaining cellulose in the solid residue. By adjusting the pretreatment conditions, a high cellulose recovery ratio of 96.5 % was achieved, with hemicellulose and lignin removal rates of 100 % and 94.7 %, respectively. Under the optimum conditions with a 1:2 M ratio of $ZrOCl_2{\cdot}8H_2O$:EG, 110 °C and 3 h, the glucose yield obtained by enzymatic hydrolysis of the cellulose reached 94.4 %. The enzymatic hydrolysis and chemical composition properties of the solid residue from each cycle were analysed to determine a sustainable biorefinery strategy. The fundamental structural properties of the recycled DES were almost unchanged throughout the recycling cycles. More importantly, the pretreatment efficiency (delignification and enzymatic saccharification) was largely maintained after the recycling process. This study proposes a strategy for the design of metal salt-based DESs, thereby enhancing biorefinery efficiency.

2. Experimental section

2.1. Materials and chemical reagents

Raw poplar sawdust was acquired from Xiyue Lumber Mill, Xuzhou, China. The poplar sawdust was pulverized into particles (ca. 40–80 mesh). The wooden particles were dried in an oven at 105 °C overnight before use. $ZrOCl_2{\cdot}8H_2O$ (99.0 %), LiCl (99.5 %), $AlCl_3$ (97.0 %) and ethylene glycol (EG, 98.0 %) were obtained from Shanghai Aladdin Biochemical Technology Company. $ZnCl_2$ (99.0 %) was purchased from Nanjing Chemical Reagent Company. Cellulase was purchased from Novozymes Investment Company. Milled wood lignin (MWL) was prepared according to a previously established method [11].

2.2. Preparation of deep eutectic solvent

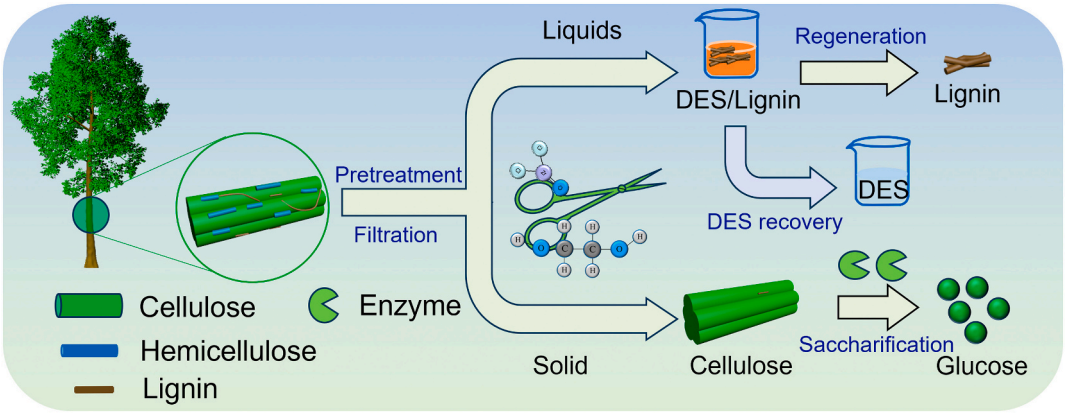
To obtain clear and uniform deep eutectic solvents, metal chlorides with different valences, including LiCl, $ZnCl_2$, $AlCl_3$, and $ZrOCl_2{\cdot}8H_2O$, were mixed with ethylene glycol at a 1:2 M ratio under stirring at 80 °C (Table S1). All solvents were stored in a desiccator for subsequent use. The Kamlet-Taft (K-T) solvatochromic parameter for DESs was calculated based on prior literature with slight modifications (details in the supporting information) [11,24].

2.3. DES pretreatment and lignin recovery

For DES pretreatment, poplar sawdust (1.0 g) was mixed with DES (20.0 g) and transferred to a glass flask (150 mL) for biomass pretreatment. Subsequently, the DES pretreatment was conducted at different temperatures (90, 100, 110, or 120 °C) for 1–4 h. After pretreatment, the mixture was separated by vacuum filtration while washing two times with a mixed solution of acetone and water (50 mL and 7:3, v/v). The solid component (residue) was collected and dried at 105 °C overnight and then stored in a desiccator. The liquid fraction was collected by rotary evaporation at 40 °C until a volume of approximately 35 mL was reached. Then, 300 mL of deionized water was added as a counter-solvent to precipitate the lignin. Lignin was recovered by vacuum filtration and then lyophilized by freeze-drying. The filtered liquid fraction was dewatered by spin evaporation to recycle the DES for the next cycle.

2.4. Material characterization

The composition of poplar sawdust and cellulose residues was analysed by National Renewable Energy Laboratory (NREL) methods (details in the supporting information). Morphological and microstructural changes in the pretreated substrates were determined using scanning electron microscopy (SEM, JSM-7600F) operated at an accelerating voltage of 5 kV. All the samples were dried and sprayed with gold before testing. Fourier transform infrared (FT-IR) spectra for the cellulose substrate and lignin were recorded using a Thermo Electron Nicolet-360 instrument. The determination of the degree of polymerization (DP) was based on the International Organization for Standardization (ISO) 5351:2012, with the specific parameters described in the supporting information. X-ray diffraction (XRD) was used to determine the degree of crystallinity of the cellulose substrate (details in the supporting information). The viscosity of the DES was measured on a Brookfield digital Viscometer DV-II at a speed of 6.0 rpm. The molecular weight (Mw) and number-average molecular weight (Mn) of lignin were determined using gel permeation chromatography (GPC, Agilent GPC/HPLC 1200) [25]. A mobile phase consisting of tetrahydrofuran was utilized at a flow rate of 1.0 mL/min. For the 2D-HSQC NMR analysis, 30 mg of lignin was weighed into a flask and sonicated with 0.6 mL of DMSO-d_6 (Sigma–Aldrich) until it formed a homogeneous liquid. A standard Bruker pulse sequence was employed for the analysis. The



Scheme 1. Illustration of the lignocellulose fractionation process by the $ZrOCl_2{\cdot}8H_2O$:EG DES.

pulse width was set to 11.27 ppm, the acquisition time was 106.8 ms, and the relaxation delay was 2 s. The pulse was acquired simultaneously with the standard Bruker pulse sequence.

2.5. Enzymatic hydrolysis

Enzymatic hydrolysis was implemented by mixing 2 g of pretreated cellulose residues and 15 FPU/g Cellic®CTec2 (150 FPU/mL) in 20 mL of citrate buffer (50 mmol/L, pH 4.8) under stirring at 200 rpm at 50 °C for 72 h. All the samples were filtered through 0.22 μm nylon syringe filters and subsequently analysed via high-performance liquid chromatography (HPLC, Agilent 1260). The conversion yield was defined as follows:

$$\text{Conversion yield (\%)} = \frac{\text{Glucose in enzymatic hydrolysis}}{\text{Initial amount of glucan unit in the poplar powder}} \times 100\% \quad (1)$$

3. Results and discussion

3.1. Pretreatment efficiency of DESs

To evaluate the pretreatment efficiency of the as-prepared DES systems, poplar sawdust was used as a feeding stock, and the components after pretreatment are summarized in Fig. 1a and b. Poplar sawdust mainly consists of cellulose (44.6 wt%), hemicellulose (16.6 wt%) and lignin (27.1 wt%). Fractionation tests were performed with a HBA (metal salt) to HBD (EG) molar ratio of 1:2 at 110 °C for 3 h. The metal salts represented by different valence states (LiCl, $ZnCl_2$, $AlCl_3$, and $ZrOCl_2{\cdot}8H_2O$) were selected as HBA, and all of them can form uniform DESs with EG (Table S1). Lignin was solubilized in DES, and hemicellulose was degraded under metal salt-induced catalytic effects [26]. The delignification efficiencies are calculated to be 14.7 % and 47.8 % for LiCl/EG and $ZnCl_2$/EG, respectively. In contrast, for the $AlCl_3$/EG DES, a noteworthy increase in delignification and hemicellulose removal is observed. However, the cellulose retention rate with $AlCl_3$/EG is only 72.7 % (Fig. 1a and b), indicating that the cellulose is degraded to some extent. This suggests that $AlCl_3$ can serve as a catalyst for cellulose depolymerization and degradation [27]. The $ZrOCl_2$/EG system has a desirable delignification performance, bearing a delignification efficiency of 94.7 %. Moreover, the majority of cellulose (96.5 %) is successfully preserved. The solubilities of lignin and hemicellulose in $ZrOCl_2$/EG were also tested at 30 °C, and both were <2 %. The valence of the metal salt significantly influences the pretreatment efficiency. This might be ascribed to the different Lewis acidities and metal coordinate states in the DESs. The K-T parameters are displayed in Table 1, where the hydrogen bond acidity (α), hydrogen bond alkalinity (β) and

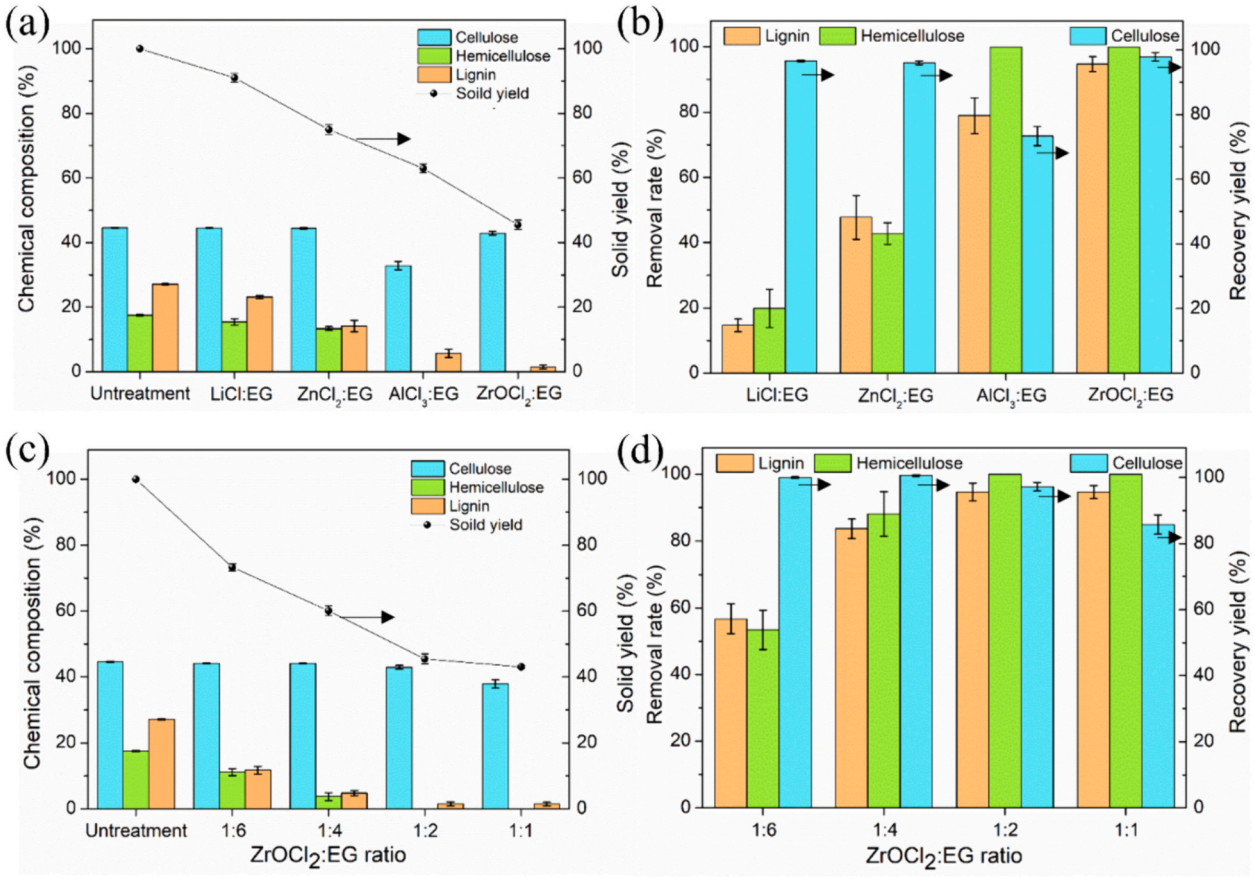


Fig. 1. Chemical compositions of poplar sawdust after pretreatment with DESs with different metal salts (a, b) and different HBA/HBD molar ratios (c, d) and the corresponding lignin and hemicellulose removal ratios and cellulose recovery yields (b, d).

Table 1
The pretreatment efficiency using different DESs and the acid properties of the as-prepared DESs.

Constituents	Molar ratio (°C)	pH	Recovery yield (%)	Removal rate (%)		K-T parameter		
			Cellulose	Hemicellulose	Lignin	π*	α	β
LiCl:EG	1:2	–	95.7	19.9	14.7	0.97	1.11	0.23
$ZnCl_2$:EG	1:2	–	95.1	42.7	47.8	1.01	1.98	0.65
$AlCl_3$:EG	1:2	–	72.7	100	78.9	1.06	2.22	0.22
$ZrOCl_2$:EG	1:1	0.1	85.0	100	94.6	1.21	2.43	0.26
$ZrOCl_2$:EG	1:2	0.8	96.5	100	94.7	1.18	2.42	0.38
$ZrOCl_2$:EG	1:4	1.1	98.9	88.1	83.8	1.14	2.17	0.38
$ZrOCl_2$:EG	1:6	2.6	99.0	53.4	56.7	1.14	2.09	0.25

polarizability (π*) are calculated. Metal ions in DESs can coordinate with EG molecules, resulting in various acid properties. The $ZrOCl_2$/EG system demonstrates the highest α and π* values, making it suitable for lignin dissolution and hemicellulose removal. Therefore, $ZrOCl_2$/EG (1:2) was selected for the following tests.

The effects of DES constituents (HBA/HBD molar ratios) on pretreatment efficiency were investigated. Compared to the untreated samples, there is a significant improvement in lignin/hemicellulose removal rates with the addition of $ZrOCl_2$:EG (Fig. 1c and d), and the pH of the DES solution decreases from 2.6 for 1:6 $ZrOCl_2$:EG to 0.1 for 1:1 $ZrOCl_2$:EG (Table 1). The $ZrOCl_2$/EG system with a molar ratio of 1:6 presents a delignification efficiency of 56.7 %. This might be attributed to the restricted accessibility of Lewis acids (metal salts) to LCC linkages. By varying the $ZrOCl_2$/EG molar ratio from 1:6 to 1:4 and 1:2, the delignification efficiency increases from 56.7 % to 83.8 % and 94.7 %, respectively. When the $ZrOCl_2$/EG molar ratio changes from 1:2 to 1:1, the delignification efficiency is calculated to be 94.6 %. However, the cellulose retention rate decreases significantly from 96.5 % to 85.0 %. This indicates that the presence of EG molecules can protect cellulose from degradation. When the amount of added EG is too low, the cellulose chains are exposed to Zr^{4+}/Cl^- ions, leading to cellulose degradation [23]. Table 1 shows that the K-T parameters of DESs are related to the HBA:HBD molar ratio. When the $ZrOCl_2$/EG molar ratio increases from 1:1 to 1:6, the α and π* values decrease from 2.43 to 2.09 and from 1.21 to 1.14, respectively. The hydroxyl groups of EG are abundant and can interact with metal salts through hydrogen bonds. The amount of EG can optimize the acidity of DESs by regulating the hydrogen bonding density and strength, realizing the effective breakage of lignin-carbohydrate bonds and retaining most of the cellulose [23].

The effects of temperature on lignin and hemicellulose removal were determined and are displayed in Fig. 2a and b [28]. When the temperature changes from 90 to 110 °C, the delignification efficiency increases from 52.7 % to 94.7 %. A higher temperature accelerates the lignin removal process because of the improved DES activity and accessibility. Additionally, the DES mobility is enhanced at higher temperatures, facilitating the exposure of DES to LCC linkages and promoting lignin removal. When the temperature further increases from 110 to 120 °C, the cellulose retention rate decreases significantly from 96.5 % to 82.6 %. This indicates that the cellulose is degraded under these relatively harsh conditions. Fig. 2c and d illustrate the effect of pretreatment time on lignin and hemicellulose removal. Extending the pretreatment duration allows DES to penetrate more thoroughly into the cellular structures of poplar, thereby enhancing the disintegration of lignin and the hydrolysis of hemicellulose. The efficiency of delignification notably increases, increasing from 61.3 % to 94.7 % as the treatment period increases from 1 to 3 h. Meanwhile, the removal of hemicellulose substantially increases, reaching 93.2 % to 100 %, respectively. However, beyond 3 h, the delignification efficiency does not increase but rather decreases slightly, which may be due to the deposition of condensed lignin and humins on the solid cellulose surface [29]. The solution

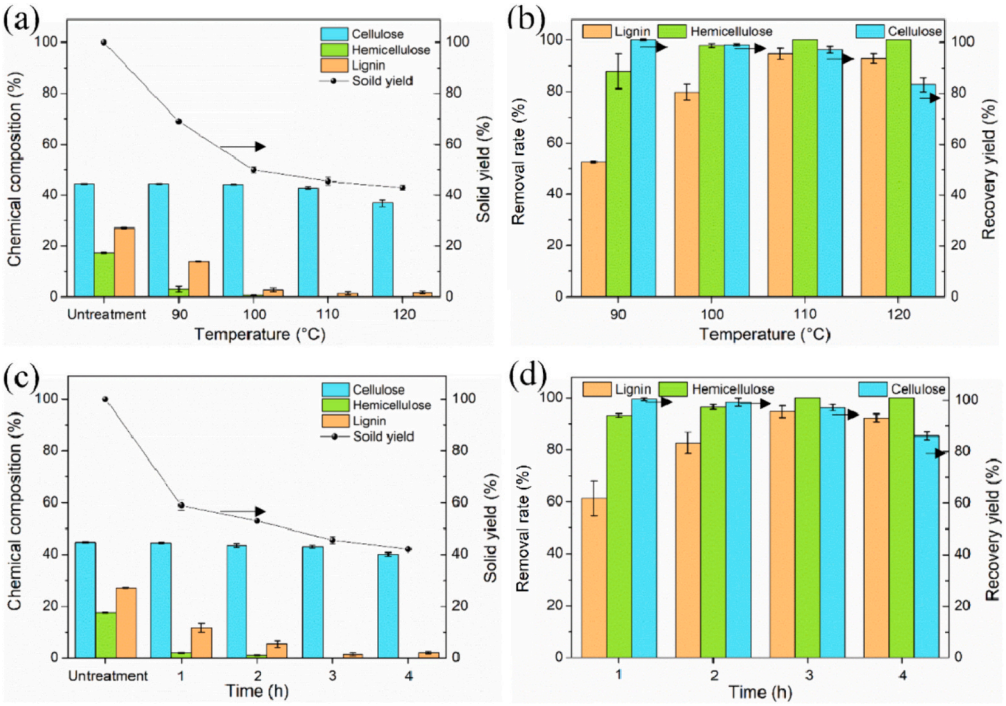


Fig. 2. The effects of pretreatment temperature (a, b) and time (c, d) on the chemical composition and lignin/hemicellulose removal rate.

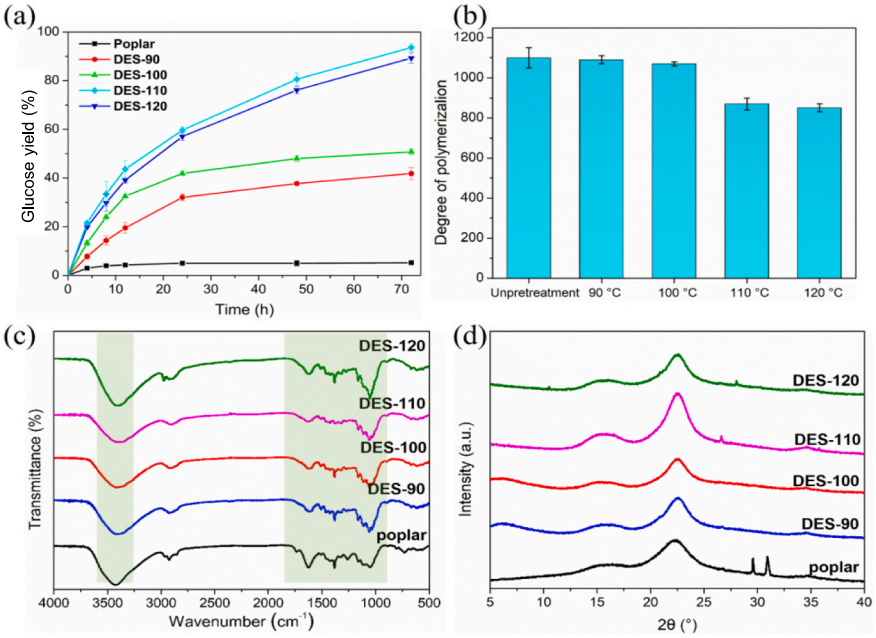


Fig. 3. Enzymatic hydrolysis (a), degree of polymerization (b), FT-IR spectra (c) and XRD patterns (d) of poplar sawdust and pretreated substrates.

composition of the DES (3 h, 110 °C) was analysed, and acetic acid, xylose, and furfural were identified at concentrations of 40, 81, and 18 mg/g, respectively (mg/g based on the weight of biomass), indicating degradation of hemicellulose during pretreatment.

3.2. Cellulose substrate characterization

Improving cellulose digestibility for glucose production is a fundamental goal in lignocellulosic biorefinery research [30]. The results showed that the glucose conversion of pristine poplar sawdust at 50 °C for 72 h is only 5.7 % (Fig. 3a). After DES pretreatment at 90 and 100 °C, the enzymatic efficiency increases to 40.9 % and 50.8 %, respectively. When the pretreatment temperature increases to 110 °C, the glucose digestibility continues to increase to 94.4 %. However, at a pretreatment temperature of 120 °C, the enzymatic efficiency decreases from 94.4 % to 89.3 %. It is possible that the pseudo-lignin formed during the pretreatment process is deposited on the cellulose surface and inhibits subsequent enzymatic digestion (Fig. S1) [11]. The performance of the optimal sample is 19.9 times greater than that of the untreated sample. This might be related to the high rate of lignin removal, which reduces the unproductive adsorption of lignin to the cellulase enzyme [21].

An in-depth examination of cellulose substrates was conducted to probe the impact of DESs on the efficiency of enzymatic digestion. The degree of polymerization of cellulose is another important characteristic that determines cellulase saccharification [31]. As shown in Fig. 3b, poplar sawdust has a DP of 1147, which is comparable to that of the majority of wood raw materials [21]. After DES pretreatment, the DP decreases significantly to 853 at 120 °C. A decrease in the DP indicates partial cleavage of the cellulose glycosidic bond, exposing more reducing ends and enhancing enzymatic saccharification [32].

FT-IR analysis (Fig. 3c) was conducted to investigate the chemical structure changes in poplar sawdust. The band at 1735 cm^{-1} is attributed to the stretching vibration of C═O in acetyl groups and carboxylic acids from hemicellulose (Fig. S2). After DES pretreatment, the peaks ascribed to C═O bonds in the spectra disappear, indicating the extraction of the hemicellulose component. The vibrations associated with the β-glycosidic bond, C—O deformation, and C—H deformation appear at 898, 1030, and 1423 cm^{-1}, respectively [32]. The characteristic peaks observed at 1513, 1541, and 1557 cm^{-1}, which correspond to the resonances of the C—C aromatic backbone inherent in lignin, have shifted noticeably or disappeared altogether [33]. This phenomenon can be ascribed to the highly effective delignification process, which enhances cellulose accessibility during subsequent enzymatic hydrolysis [34].

Table 2
Crystalline parameters of the DES-treated poplar samples.

Sample	Temperature (°C)	CrI (%)	Crystallite size (nm)	Distance between atomic layers (pm)
Untreated	–	44.8	2.63	398.1
$ZrOCl_2$:EG (1:2)	90	56.0	2.78	396.1
$ZrOCl_2$:EG (1:2)	100	56.4	2.80	397.0
$ZrOCl_2$:EG (1:2)	110	55.1	3.14	396.8
$ZrOCl_2$:EG (1:2)	120	52.4	3.15	396.4

Fig. 3d shows the crystal structure of the pretreated poplar samples. The crystallinity index (CrI), crystal size (L) and distance between atomic layers (D) of the pristine and pretreated substrates were determined. As shown in Table 2, the CrI of the raw material is 44.8 %, while the CrI of the pretreated substrate increases to 56.4 % at 100 °C. This increasing trend indicates that amorphous hemicellulose and lignin are removed during pretreatment, which is consistent with the results of the component analysis. However, CrI decreases at temperatures of 110 and 120 °C, which is due to the decrystallization effect of the DES treatment [24].

The size of the hydrophobic lattice planes (200) in crystalline cellulose is considered crucial for cellulase adsorption [24]. According to the observations presented in Table 2, the D spacing (the interatomic distance between layers) of the cellulose remains relatively consistent. However, significant increases in the crystallite dimensions of the pretreated cellulose are observed, especially at temperatures of 110 and 120 °C. The crystal sizes reach 3.14 and 3.15 nm, aligning with a discernible enhancement in the saccharification efficiency [35].

SEM was employed to observe the changes in the surface morphology and microstructure of the poplar sawdust before and after pretreatment.

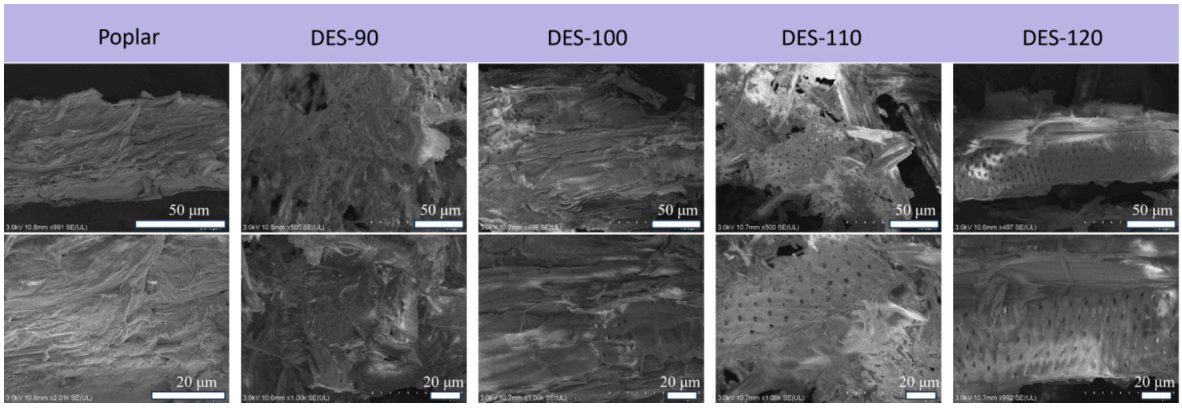


Fig. 4. SEM images of cellulose substrates pretreated with DES($ZrOCl_2$-EG) at different temperatures.

Fig. 4 shows that the original matrix has a smooth and rigid morphology, potentially impeding enzyme access to cellulose [21]. The surface of the matrix becomes rougher after DES pretreatment. Notably, as the pretreatment temperature increases, many pores appear in the cellulose substrate. All these changes result in the creation of a large number of reaction sites on the fibre surface, which increases the accessibility of the enzyme and improves the efficiency of enzymatic saccharification.

3.3. Lignin characterization

Lignin primarily comprises the following phenylpropanoid units: syringyl (S), guaiacyl (G), and *p*-hydroxyphenyl (H), which are interconnected through ether and carbon–carbon bonds. Under optimal pretreatment conditions, the lignin purity reaches 97.1 %. Structural changes in lignin were further analysed by FT-IR analysis of the recovered lignin (Fig. S3) [36]. The effect of temperature on the molecular weight and structural properties of the poplar samples was investigated using GPC. The regenerated lignin at 110 °C has a molecular weight of 1501 g/mol and a notably low polydispersity index of 1.7 (Table S2). This suggests that the lignin produced is uniform and homogeneous. This result strongly indicates that acidic DESs effectively promote the dissociation and depolymerization of lignin. Depolymerization mainly involves the cleavage of β-O-4 linkages, and the $ZrOCl_2$-EG DES is effective in providing sufficient protons to accelerate this complex process. It is presumed that EG molecules do not have a high pretreatment capacity for lignocellulose. The combination of $ZrOCl_2$ and EG effectively isolates lignocellulose, and the molecular weight of the recovered lignin decreases as the temperature increases.

2D-HSQC-NMR spectroscopy was used to elucidate the dissolution mechanism of lignin (Fig. 5 and Table S3) [25]. MWL sourced from poplar sawdust was used as a control sample to provide a definitive reference, enabling clear identification and annotation of the cross peaks. By segregating the 2D NMR spectra into aromatic (δ_C/δ_H = 160–90 ppm/8.0–5.5 ppm) and side-chain (δ_C/δ_H = 95–45 ppm/5.5–2.5 ppm) regions, a comprehensive analysis was achieved. For the MWL, the presence of β-O-4 linkages is notable. Conversely, the β-O-4

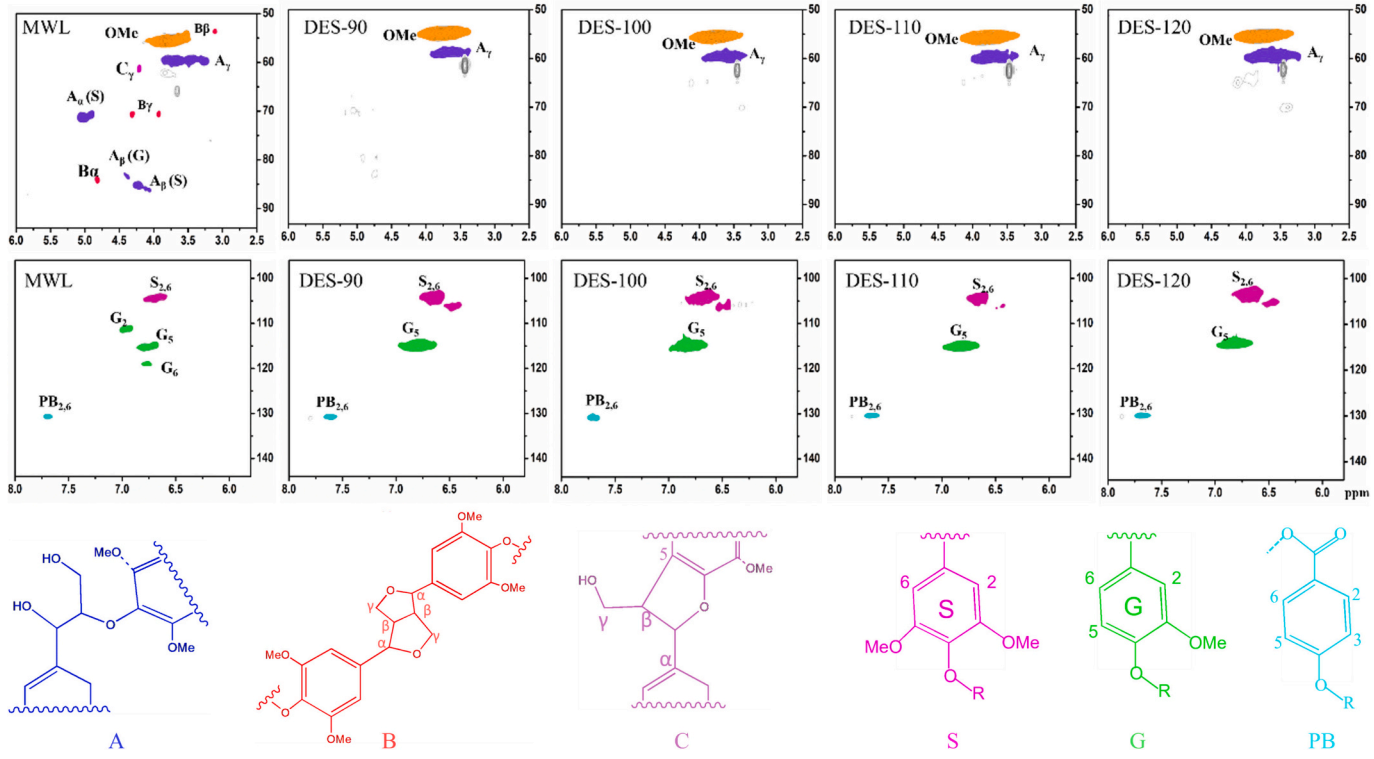


Fig. 5. Side-chain regions and aromatic regions in the 2D-HSQC NMR spectra of lignin samples.

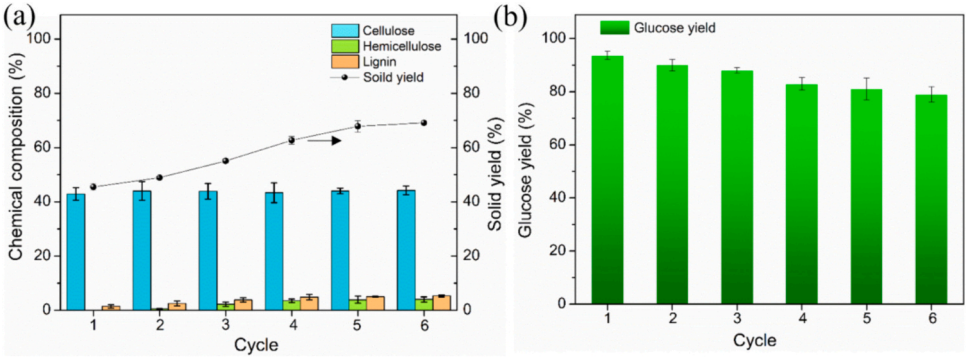


Fig. 6. Characterization of cellulose substrates in consecutive 6 cycles: (a) chemical composition and solid yield and (b) glucose yield after enzymatic hydrolysis.

bond signal decreases remarkably in the DES-extracted lignin [26]. In addition, the aromatic region of the isolated lignin shows obvious signals corresponding to the guaiacyl (G) and syringyl (S) units (Table S4) [37].

3.4. Cycle tests

The recovery of DESs is critically important for assessing the economic viability of biomass refinement [38]. Fig. 6a illustrates the solid recovery and chemical composition of the cellulose substrate across 6 consecutive cycles. Each cycle shows an increase in solid yield and remarkable cellulose retention (>90 %). The recovery of the DES solution at the end of each cycle was >95 %, suggesting its high recyclability (Fig. S4). The utilization of the $ZrOCl_2$-EG DES in the process highlights its ability to achieve effective closed-loop recovery and reuse of solvents without imposing significant environmental burdens or costs. Fig. 6b illustrates a slight decrease in enzymatic hydrolysis performance following 6 cycles, probably due to the introduction of some impurities in the recycled DES (with an increased viscosity of 72.1 mPa·s and pH of 0.9 after 6 cycles of DES). On the basis of these results, Zr-based DES pretreatment has great potential for biorefineries and for the development of biofuels [39].

4. Conclusions

This study describes a Zr-based DES system comprising $ZrOCl_2{\cdot}8H_2O$ and EG for poplar sawdust pretreatment. The Zr^{4+} ions in the salt provide Lewis acidity, which is essential for lignocellulose processing. The results indicate that DES pretreatment significantly eliminates hemicelluloses and lignin, thereby significantly reducing biomass recalcitrance. Under the optimum conditions with a 1:2 M ratio of $ZrOCl_2{\cdot}8H_2O$:EG, 110 °C and 3 h, the glucose yield obtained by enzymatic hydrolysis reached 94.4 %, which was significantly greater by 19.9 times than that of the original biomass without DES pretreatment. The molecular weight of the DES-extracted lignin was 1501 g/mol, with a uniform distribution and a polydispersity of 1.7. More importantly, the pretreatment efficiency (delignification and enzymatic saccharification) was largely maintained after the recycling process. This study presented an emerging method for the effective fractionation and valorization of lignocellulosic biomass within biorefinery technology.

CRediT authorship contribution statement

Yunhua Bai: Writing – original draft, Investigation, Formal analysis. **Xiong-Fei Zhang:** Writing – original draft, Supervision, Formal analysis, Conceptualization. **Mengjiao Yu:** Investigation, Formal analysis. **Jianfeng Yao:** Writing – review & editing, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

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Appendix A. Supplementary data

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